UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

*3236105264*

**CHEMISTRY** **5070/03**

Paper 3 Practical Test **October/November 2009**

**1 hour 30 minutes**

Candidates answer on the Question Paper

Additional Materials: As listed in the Instructions to Supervisors.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Qualitative Analysis Notes are printed on page 8.
You should show the essential steps in any calculations and record experimental results in the spaces provided on the question paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| 1 | |
| 2 | |
| Total | |

This document consists of **7** printed pages and **1** blank page.


© UCLES 2009

*For Examiner's Use*

**1** **P** is a solution containing an impure sample of sodium carbonate. The impurity is another sodium compound. You are to identify the impurity and then determine, by titration with hydrochloric acid, the amount of sodium carbonate present in the impure sample.

**P** contains 6.00 g of the impure sodium carbonate in 1.00 $dm^3$.
**Q** is 0.100 mol/$dm^3$ hydrochloric acid.

**(a)** Identification of the impurity in **P**.

Carry out the following tests on solution **P** and record your observations in the table.

| test no. | test | observations |
|---|---|---|
| **1** | **(a)** To a portion of **P**, add an equal volume of aqueous barium nitrate.<br><br>**(b)** Add dilute nitric acid to the mixture from **(a)**. | |
| **2** | **(a)** To a portion of **P**, add a few drops of aqueous silver nitrate.<br><br>**(b)** Add dilute nitric acid to the mixture from **(a)**. | |

The formula of the impurity present in **P** is ..... [5]

© UCLES 2009

**(b)** Determination of the concentration of the sodium carbonate in **P**.

*For Examiner's Use*

Put **Q** into the burette.

Pipette a 25.0 $cm^3$ (or 20.0 $cm^3$) portion of **P** into a flask and titrate with **Q**, using the indicator provided.

Record your results in the table, repeating the titration as many times as you consider necessary to achieve consistent results.

**Results**

*Burette readings*

| titration number | 1 | 2 | |
|---|---|---|---|
| final reading / $cm^3$ | | | |
| initial reading / $cm^3$ | | | |
| volume of **Q** used / $cm^3$ | | | |
| best titration results (✓) | | | |

**Summary**

Tick (✓) the best titration results.

Using these results, the average volume of **Q** required was .....$cm^3$.

Volume of solution **P** used was .....$cm^3$.

[12]

**(c)** **Q** is 0.100 mol/dm$^3$ hydrochloric acid.

*For Examiner's Use*

Using your results from **(b)**, calculate the concentration, in mol/dm$^3$, of the sodium carbonate in **P**.

$$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$$

concentration of sodium carbonate in **P** ..... mol/dm$^3$. [2]

**(d)** Using your answer from **(c)**, calculate the mass of sodium carbonate present in 1.00 dm$^3$ of solution **P**.
[The relative formula mass of sodium carbonate is 106.]

mass of sodium carbonate present in 1.00 dm$^3$ of solution **P**.....g [1]

**(e)** Using your answer from **(d)**, calculate the percentage by mass of sodium carbonate in the impure sample.

percentage by mass of sodium carbonate in the impure sample ..... [1]

[Total: 21]

© UCLES 2009

**BLANK PAGE**

**2** You are provided with three solutions **R**, **S**, and **T**. Carry out the following tests and record your observations in the table. You should test and name any gas evolved.

*For Examiner's Use*

| test no. | test | observations with solution **R** |
|---|---|---|
| **1** | **(a)** To a portion of the solution, add aqueous ammonia until a change is seen.<br><br>**(b)** Add excess aqueous ammonia to the mixture from **(a)**. | |
| **2** | **(a)** To a portion of the solution, add aqueous sodium hydroxide until a change is seen.<br><br>**(b)** Add excess aqueous sodium hydroxide to the mixture from **(a)**.<br><br>**(c)** Add aqueous hydrogen peroxide to the mixture from **(b)**. | |
| **3** | To a portion of the solution, add an equal volume of dilute sulfuric acid and then add a few drops of aqueous potassium manganate(VII). | |

© UCLES 2009

| observations with solution **S** | observations with solution **T** |
|---|---|
| | |
| | |
| | |

[17]

*For Examiner's Use*

**Conclusion**

The formula of the cation present in solution **R** is .....

The formula of the cation present in solution **S** is ..... [2]

[Total: 19]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

© UCLES 2009

## QUALITATIVE ANALYSIS NOTES

### Tests for anions

| *anion* | *test* | *test result* |
|---|---|---|
| carbonate ($CO_3^{2-}$) | add dilute acid | effervescence, carbon dioxide produced |
| chloride ($Cl^-$) [in solution] | acidify with dilute nitric acid, then add aqueous silver nitrate | white ppt. |
| iodine ($I^-$) [in solution] | acidify with dilute nitric acid, then add aqueous lead(II) nitrate | yellow ppt. |
| nitrate ($NO_3^-$) [in solution] | add aqueous sodium hydroxide, then add aluminium foil; warm carefully | ammonia produced |
| sulfate ($SO_4^{2-}$) [in solution] | acidify with dilute nitric acid, then add aqueous barium nitrate | white ppt. |

### Tests for aqueous cations

| *cation* | *effect of aqueous sodium hydroxide* | *effect of aqueous ammonia* |
|---|---|---|
| aluminium ($Al^{3+}$) | white ppt., soluble in excess giving a colourless solution | white ppt., insoluble in excess |
| ammonium ($NH_4^+$) | ammonia produced on warming | – |
| calcium ($Ca^{2+}$) | white ppt., insoluble in excess | no ppt., or very slight white ppt. |
| copper ($Cu^{2+}$) | light blue ppt., insoluble in excess | light blue ppt., soluble in excess giving a dark blue solution |
| iron(II) ($Fe^{2+}$) | green ppt., insoluble in excess | green ppt., insoluble in excess |
| iron(III) ($Fe^{3+}$) | red-brown ppt., insoluble in excess | red-brown ppt., insoluble in excess |
| zinc ($Zn^{2+}$) | white ppt., soluble in excess giving a colourless solution | white ppt., soluble in excess giving a colourless solution |

### Tests for gases

| *gas* | *test and test result* |
|---|---|
| ammonia ($NH_3$) | turns damp red litmus paper blue |
| carbon dioxide ($CO_2$) | turns limewater milky |
| chlorine ($Cl_2$) | bleaches damp litmus paper |
| hydrogen ($H_2$) | ‘pops’ with a lighted splint |
| oxygen ($O_2$) | relights a glowing splint |
| sulfur dioxide ($SO_2$) | turns aqueous potassium dichromate(VI) from orange to green |

© UCLES 2009